Sums, Quotients, and Complexes

Math 5250 — Week 2

Direct Sums

Let V, W be vector spaces over a field $\mathbf{F}$. The **direct sum**

$$V \oplus W = \{(v, w) : v \in V,\ w \in W\}$$

is a vector space with $(v, w) + (v', w') = (v + v', w + w')$ and $a(v, w) = (av, aw)$.

- The direct sum of a finite family $\{V_i\}_{i=1}^m$ is the set of sequences $(v_1, \ldots, v_m)$, $v_i \in V_i$.
- For an index set I, $\oplus_{i \in I} V_i$ is the set of $f : I \to \bigcup_i V_i$ with $f(i) \in V_i$, nonzero for only finitely many i.
- $\mathbf{F}$ is a one-dimensional vector space over itself, and $\mathbf{F}^n = \oplus_{i=1}^n \mathbf{F}$.

Direct Sums: Dimension

V and W sit inside $V \oplus W$ as $(v, 0)$ and $(0, w)$, intersecting only in $(0, 0)$.

Proposition. If V, W are finite dimensional of dimensions n, m, then $V \oplus W$ is finite dimensional of dimension $n + m$; if $v_1, \ldots, v_n$ is a basis of V and $w_1, \ldots, w_m$ a basis of W, then

$$(v_1, 0), \ldots, (v_n, 0), (0, w_1), \ldots, (0, w_m)$$

is a basis of $V \oplus W$.

Subspace Sums

Let U, W be subspaces of V. The **sum**

$$U + W = \{u + w : u \in U,\ w \in W\}$$

is a subspace of V.

- It is the smallest subspace of V containing both U and W.
- If $u_1, \ldots, u_n$ span U and $w_1, \ldots, w_k$ span W, then $u_1, \ldots, u_n, w_1, \ldots, w_k$ span $U + W$.
- $\dim(U + W) \leq \dim(U) + \dim(W)$, and this can be strict (e.g. $U = W$).

Algorithm to Compute a Basis for $U + W$

1. Choose a basis for V and spanning sets for U and W as above. Form the matrix A whose columns are the u_i and w_j in the chosen basis.
2. Row-reduce A to A'.
3. The columns of A corresponding to pivot columns of A' form a basis for $U + W$.
4. This basis is a subset of the union of the u_i and w_j.

Internal Direct Sums

Definition. If every $x \in U + W$ has a *unique* representation $x = u + w$ with $u \in U$, $w \in W$, then $U + W$ is an (internal) direct sum, isomorphic to $U \oplus W$.

Proposition. $U + W$ is a direct sum if and only if either:

a. $U \cap W = 0$, or
b. $\dim(U) + \dim(W) = \dim(U + W)$.

A Dimension Formula

Proposition. For subspaces U, W of V,

$$\dim(U+W) = \dim(U\oplus W) - \dim(U\cap W) = \dim(U) + \dim(W) - \dim(U\cap$$

Equivalence Relations

Definition. An *equivalence relation* on X is a relation $x \sim y$ satisfying:

1. $x \sim x$ for all x.
2. $x \sim y \implies y \sim x$.
3. $x \sim y,\ y \sim z \implies x \sim z$.

An equivalence relation partitions X into equivalence classes $[x] = \{y \in X : y \sim x\}$; distinct classes are disjoint.

Quotient Spaces

Let V be a vector space over F and $W \subset V$ a subspace. Define $x \sim y \iff x - y \in W$.

Definition. The set of equivalence classes $[x]$ is the **quotient space** V/W, a vector space over F with

$$[x] + [y] = [x + y], \qquad a[x] = [ax].$$

Quotient Spaces: Dimension

Proposition. If V is finite dimensional and $W \subset V$, then

$$\dim(V/W) = \dim(V) - \dim(W).$$

(Via the linear map $f : V \to V/W$, $f(v) = [v]$, which is onto with $\mathrm{null}(f) = W$.)

Universal Property of Quotients

Proposition. Let $f : V \to W$ be linear and $H \subset V$ contained in $\ker(f)$. Then there is a unique linear map $\overline{f} : V/H \to W$ with $\overline{f}([v]) = f(v)$ for all $v \in V$.

The subspaces of V/H correspond exactly to the subspaces of V containing H.

Complexes

Definition. A *complex* of vector spaces is a sequence W^i with linear maps

$$\cdots \to W^i \xrightarrow{d_i} W^{i+1} \xrightarrow{d_{i+1}} W^{i+2} \to \cdots$$

satisfying $d_{i+1}d_i = 0$ for all i. The complex is *bounded* if all but finitely many W^i are zero.

Let $Z^i = \ker(d_i)$ (the **cocycles**) and $B^i = d_{i-1}(W^{i-1})$ (the **coboundaries**). The i^{th} **cohomology** is $H^i(W^\cdot) = Z^i/B^i$.

Euler Characteristic

For a bounded complex of finite dimensional spaces, the **Euler characteristic** is

$$\chi(W^{\cdot}) = \sum_i (-1)^i \dim(W^i).$$

Proposition.

$$\chi(W^{\cdot}) = \sum_i (-1)^i \dim H^i(W^{\cdot}).$$

Exact Complexes

Definition. A complex is **exact** if all its cohomology groups are zero; equivalently $\ker(d^i) = \mathrm{image}(d^{i-1})$ for all i.

A **short exact sequence** is a 5-term exact complex

$$0 \to A \to B \to C \to 0,$$

so $A \to B$ is injective, $B \to C$ is surjective, and $C \cong A/B$.

Homology

For a complex indexed the other way, $d_i : W_i \to W_{i-1}$:

- $Z_i = \ker(d_i)$ is the space of *i*-**cycles**.
- $B_i = \mathrm{image}(d_{i+1})$ is the space of *i*-**boundaries**.
- $H_i(W^\cdot) = Z_i / B_i$ is the i^{th} **homology group**.

Maps of Complexes

Definition. A map of complexes $f : \{W^\cdot\} \to \{V^\cdot\}$ is a family $f_i : W^i \to V^i$ with $f_{i+1} \circ d_i = d_i \circ f_i$.

Such an f carries cocycles to cocycles and coboundaries to coboundaries, and therefore induces the **induced map on cohomology**

$$\overline{f} : H^i(W^\cdot) \to H^i(V^\cdot).$$